PalmSource™ 99

Developing the Wireless World



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3Com
More connected.

Agenda

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- Who is JP Systems?
- Our Vision of a Wireless World
- One-Touch Messaging Platform
- InfoBeam
- Summary
- Glenayre Discussion
- Q&A

JP Systems Mission

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Provide innovative and easy to use software products and solutions for anywhere, anytime data access using wireless handheld terminals



JP Systems Vision— Simplifying Wireless Data

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Client-Based Products

Wireless Devices: PDAs, Phones, Pagers

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Server-Based Products

Internet Content

Remote Mail

Scheduling

Database

Transactions

Corporate Data

JP Systems provides Server- and Client-based products to simplify wireless connectivity

Wireless Data Today

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- Networks
  - 2-Way Paging
  - Mobitex
  - CDPD
  - GSM
  - Ricochet
- Smart Phones
  - WAP enabled
- Hand-held Data Devices
  - Palm OS® devices

Integration Today

- Smart Phones
  - WAP
- Palm VII devices
  - US Only, Bell South
- Handspring
  - Various modules





Integration vs. Companionship

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- Why did you buy your Palm Computing® Platform device?
- Why did you buy your phone?

- Do you really want them together???

- A companion strategy allows you to use the appropriate platform for the data being accessed

Beaming Information

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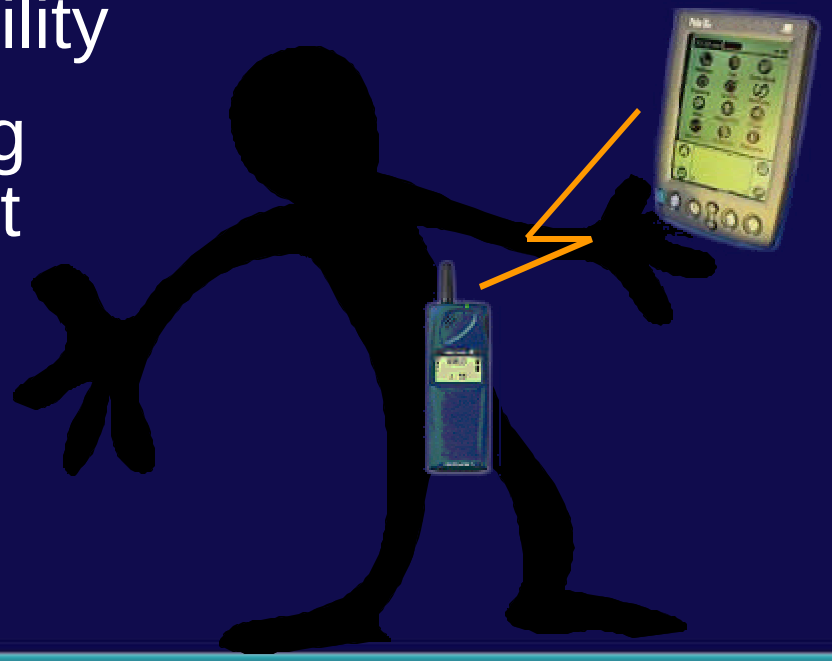
- Here today
- Use 2-way pagers and phones as modems
- Ready for Bluetooth!



Bluetooth Cometh!

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- Bluetooth enabled devices will provide the glue for connecting “companions”
- Users won’t be tied to small screens to read WAP information
- Convenience and portability
- JP Systems is supporting Widcomm’s BlueConnect



One-Touch Messaging Platform



Wireless Modem



Handspring Modem



WidComm Bluetooth





Nokia 7100



Glenayre AccessLink II

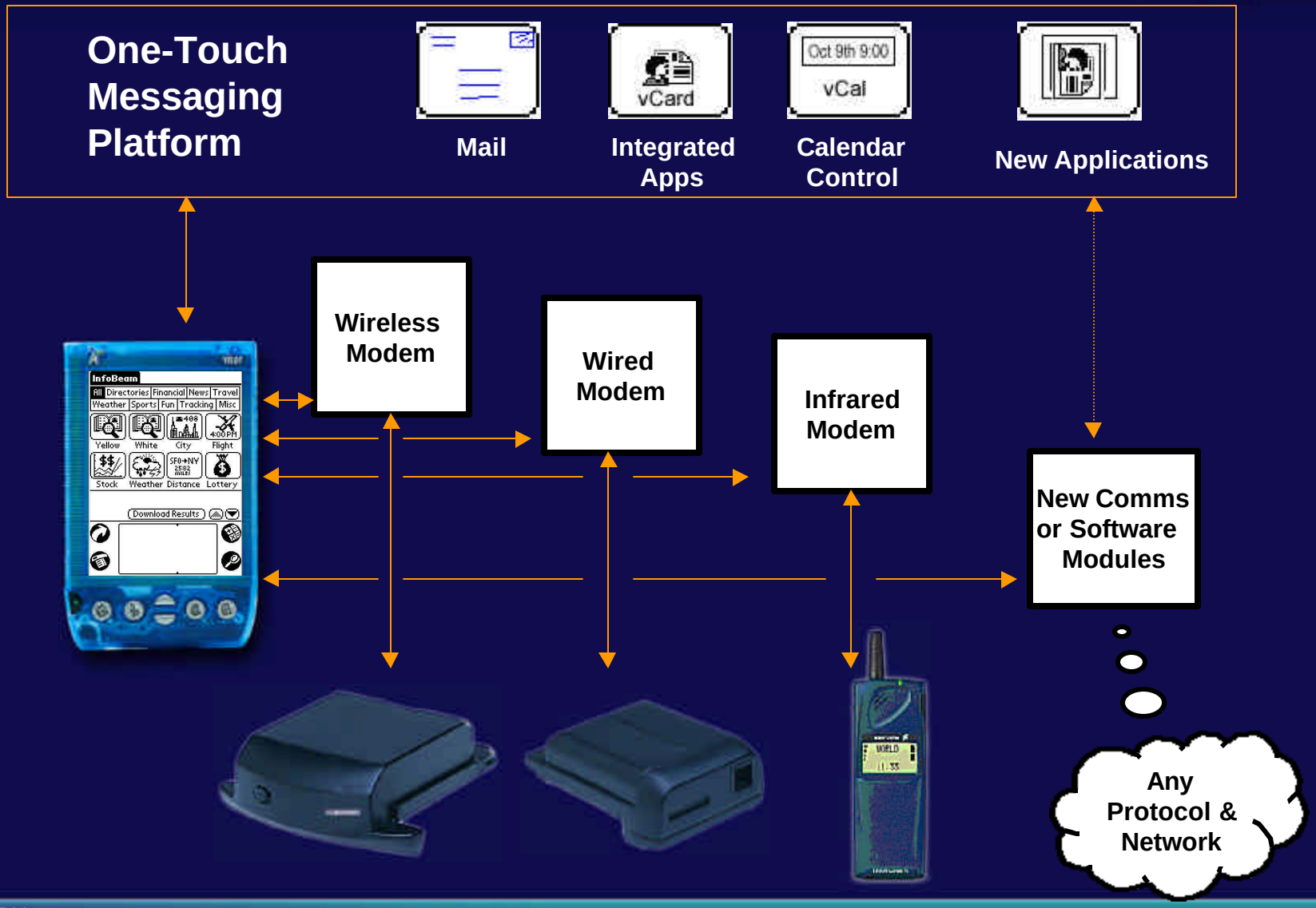


Outlook Synchronization

Plug'n'Play Messaging Platform for Palm OS

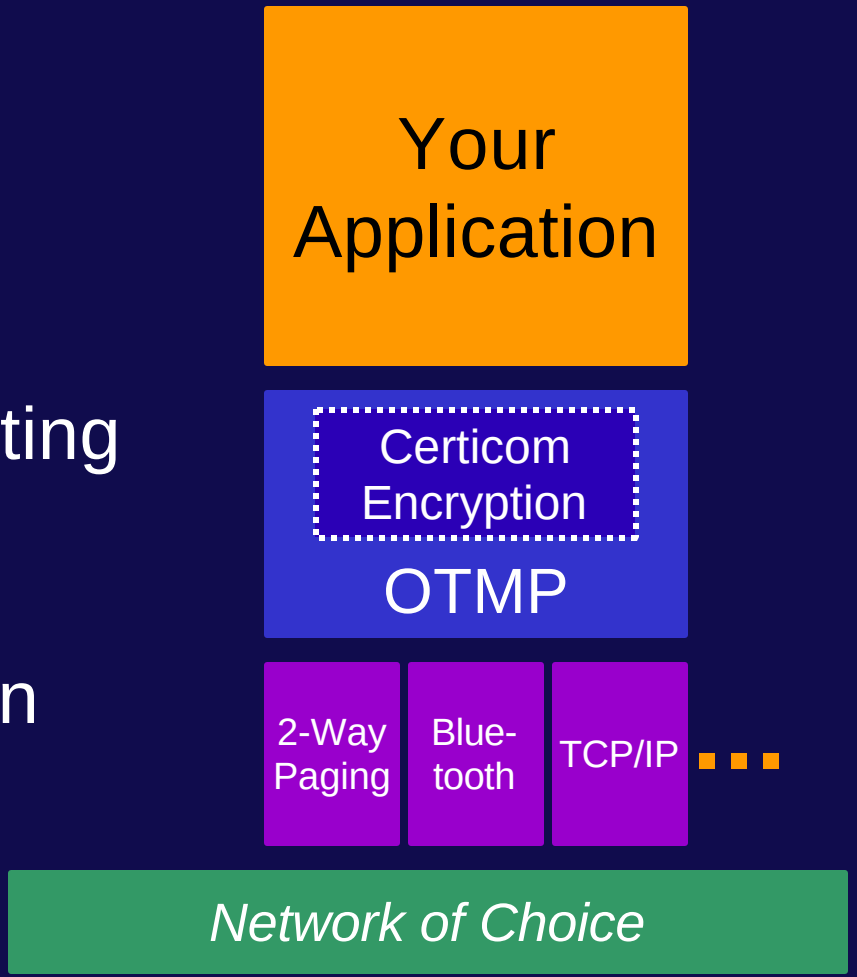
Plug-in Architecture for Handspring

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Wireless SDK Concept

- Advanced Message Architecture
- Provides a unified, consistent interface
- Plug-and-play with existing applications
- Optional Certicom's Elliptic Curve Encryption



Developing Messaging Apps

- Using OTMP SDK
  - Acquire from JP Systems
  - Sign-up at www.jpsystems.com/developer
- Develop advanced messaging apps
  - Create apps that include support for remote calendaring, beaming, adding attachments
  - Examples: In-home patient care, on-site order procession, on-site inventory, order tracking, etc.
- Message router—each application can receive messages intended for them
- Implementing Certicom Elliptic Curve Security

Application Versatility

- Same application can work with local area or wide area networks
  - Bluetooth
  - 2-Way Paging
  - GSM
  - Wireline Modem

Wireless Internet Content

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Concept

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- Wireless Connectivity to Internet/Intranet Content
  - Universal Web content delivery
  - Enterprise solution support
- Server and Client
  - Tools to develop custom solutions
- Uses OTMP transports

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Wireless Information Services



- InfoBeam provides multiple services on demand
- Each Icon on the screen shown represents a service
- Service sampling:
  - Yellow Pages
  - White Pages
  - Flight Status
  - Driving Directions
  - Stock Quotes
  - Restaurant Search
  - FedEx / UPS package tracking

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User Experience



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Upcoming Services



- BarPoint.com
  - Competitive Pricing
  - Ability to order product on-line from major e-commerce sites from your handheld device
- Airline notification
  - Gate Info
  - Flight reservations
  - Cancellation notifications
- Traffic conditions
- Realtor links
  - Providing MLS listings on-demand

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InfoBeam Service Process

Example: Calendar Request

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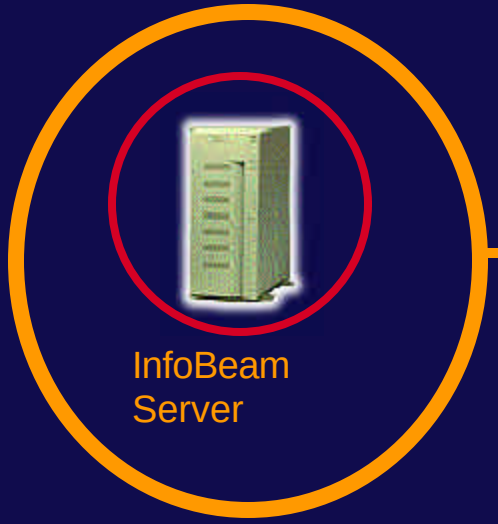




E-mail

Intranet

Group Calendar



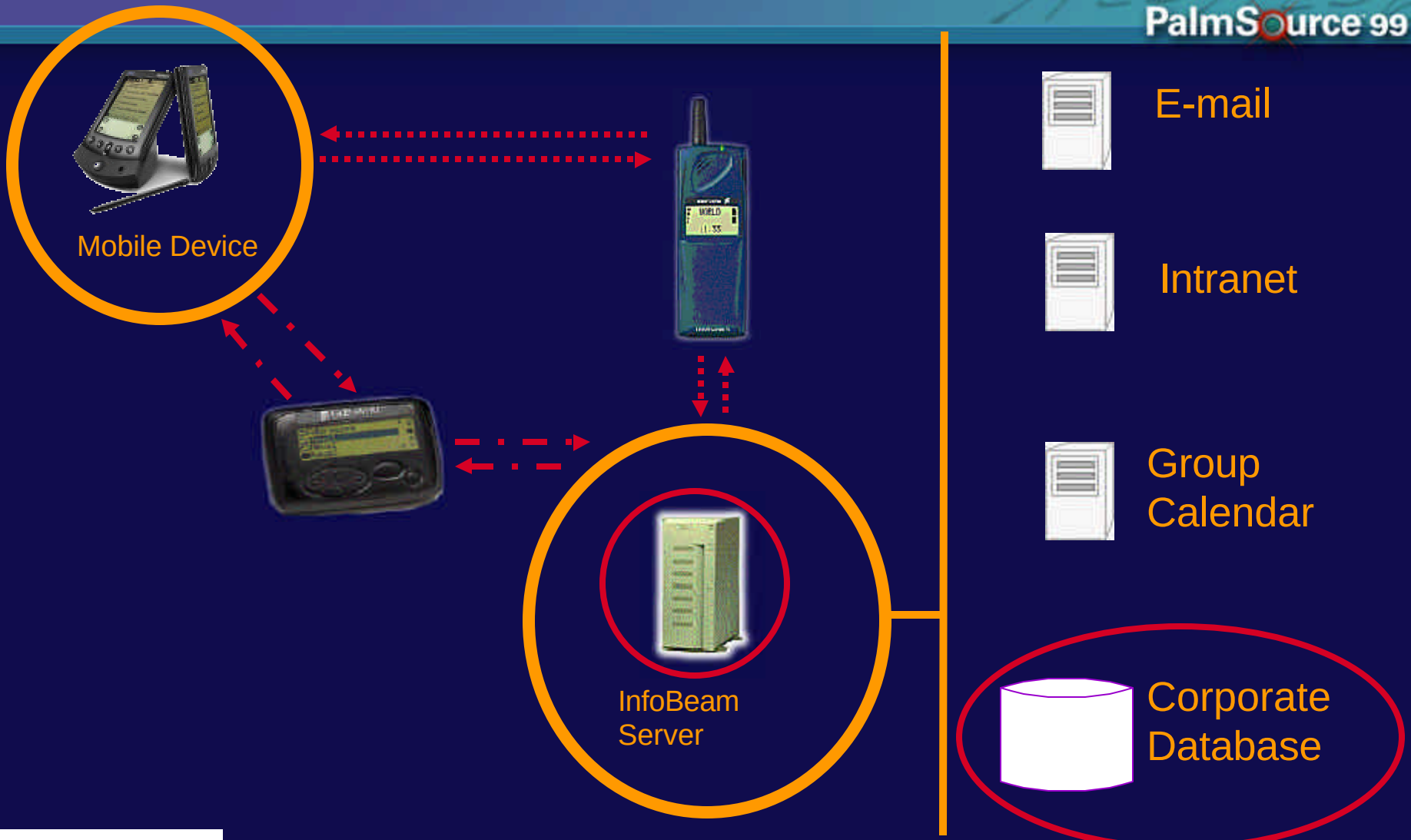
Corporate Database

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InfoBeam Service Process

Example: Database Transaction

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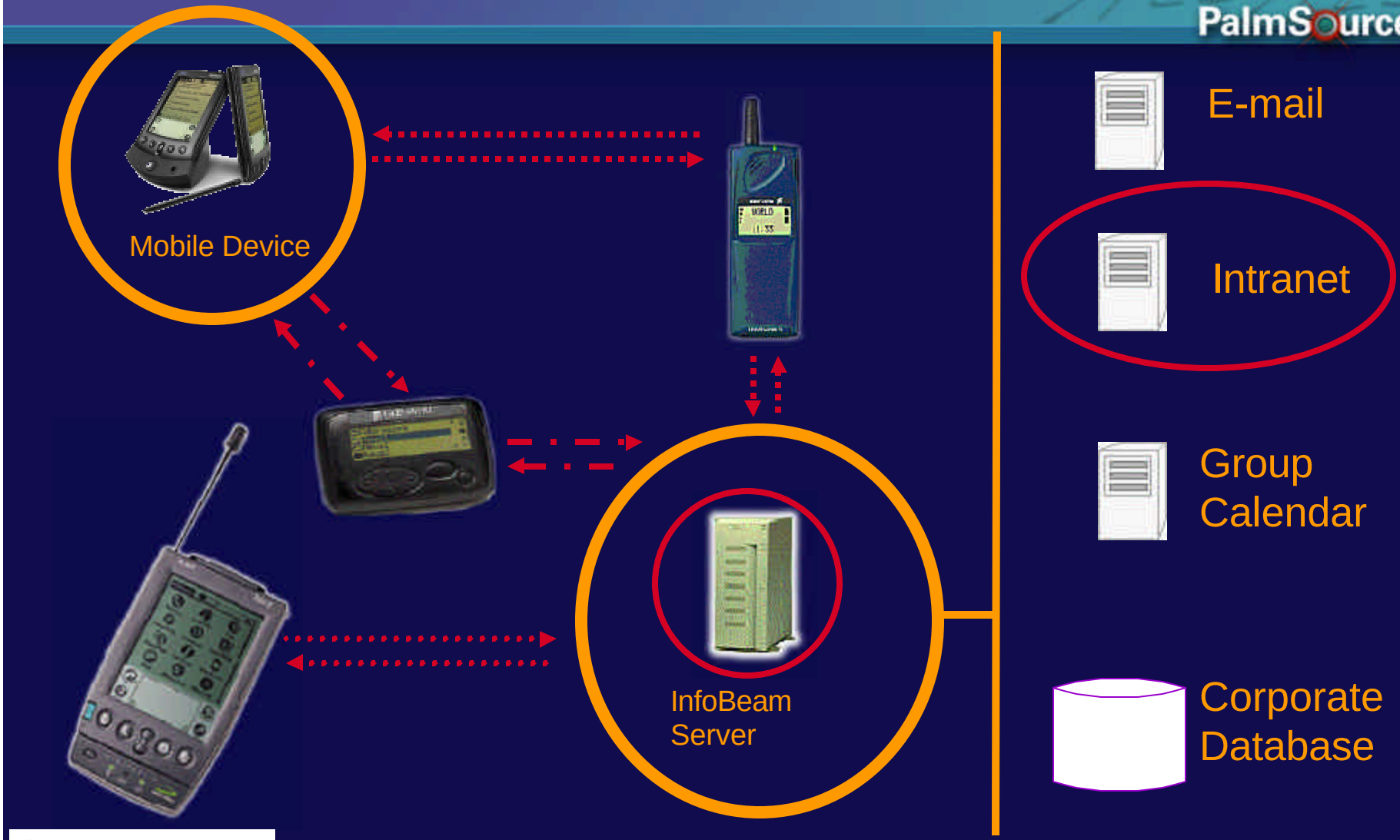


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InfoBeam Service Process

Example: Intranet Information

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Scalable Design

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- Available for any data-ready network
- Modular design for “plug-in” communication components such as:
  - Current:
    - GSM phones
    - CDPD
    - SMS
    - Two-Way Pagers
    - Wireline Modems
  - Future:
    - GPRS
    - WAP

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Developing / Deploying

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- Content such as BarPoint.com
- Adding services—new level
- Enterprise solutions—new screen, stock, price list, document faxing, ERP

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Summary

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- The Enterprise needs wireless solutions today
- Solutions come in various configurations
  - Companions
  - Integrated
- InfoBeam and One-Touch Mail can be the foundations for integrating wireless support into your applications
- JP Systems' Wireless SDK is available now
- Visit us online at www.jpsystems.com

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ReFLEX—
The Future of Wireless Data?

Kin Seto



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What Is ReFLEX?

- ReFLEX is a protocol for data communication over radio links that is part of the FLEX family of protocols, created by Motorola for the paging industry
- FLEX is a protocol that allows one-way broadcast from radio base stations to pagers
- ReFLEX adds the component that carries data back from the device, which makes it the basis for a fully two-way wireless data system



ReFLEX Advantages

- ReFLEX has the best coverage and more networks than any digital radio system in the US
- Low cost, small size, and long battery life devices
- The best radio reception, including inside buildings
- Low cost, low risk, easy to implement applications
- Great potential for new Wireless Messaging Applications



ReFLEX Basics

- High-power, simulcast or targeted network
- 6400 bps to device
- 800, 1600, 6400, or 9600 bps from device
- Automatic roaming
  - Nation-wide coverage with 1-way paging economics
- Messages can be guaranteed with “ACK”
- Normal messages can be sent as ASCII text
- Special applications can be sent as binary messages

Successful ReFLEX Applications:

- Are easy and/or fun to use
- Make money or save money for someone
- Are insensitive to latency or speed considerations
- Take advantage of ReFLEX's special features:
  - Broadcast
  - Multiple-choice responses
  - Information "push" model
  - Assure delivery
  - Full two-way capabilities
  - Wireless e-mail
  - PDA-pager companion strategy

Glenayre Developer's Program

- Glenayre Consumer Products Group based in Santa Clara, CA is actively looking for a limited number of developers for the Glenayre Preferred Developers Program
- The preferred developers program focuses on developers that can bring new applications for Palm Computing® Platform
- Specific applications:
  - E-mail
  - Transaction-based: banking, stock purchase, airline
  - Ticket purchase, etc
  - Web clipping: access to bank account balance,
  - Driving directions, flight info, stock quotes, etc.
  - Intranet

Glenayre Developer's Program (Cont.)

- Specific Applications (cont.)
  - Financial: access to personal portfolio or bank accounts
  - Data gathering
  - Specific vertical applications
- Glenayre Contact Person(s)
  - **Kin Seto**
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